STAT293: Tutorial 1 Solutions

1. (a) $E(V) = E(Y_1) - 3E(Y_2) + E(Y_3) = 25 - 3(3.8) - 5.3 = 8.3$
   (b) $Var(V) = Var(Y_1) + 3^2 * Var(Y_2) + Var(Y_3) = 10 + 9(10) + 4 = 104$
   (c) $E(Y_2Y_3) = E(Y_2)E(Y_3) = 3.8 \times (-5.3) = -20.14$

2. (a)

$$\begin{aligned} E\left(\sum_{i=1}^{5} X_i\right) &= \sum_{i=1}^{5} E(X_i) \\ &= \sum_{i=1}^{5} \mu_i \end{aligned}$$

   (b)

$$\begin{aligned} E\left(\bar{X}\right) &= E\left(\sum_{i=1}^{5} \frac{1}{5} X_i\right) \\ &= \frac{1}{5} \sum_{i=1}^{5} E(X_i) \\ &= \frac{1}{5} \sum_{i=1}^{5} \mu_i \end{aligned}$$

   (c)

$$\begin{aligned} Var\left(\sum_{i=1}^{5} X_i\right) &= \sum_{i=1}^{5} Var(X_i) \\ &= \sum_{i=1}^{5} \sigma^2 \\ &= 5\sigma^2 \end{aligned}$$

   (d)

$$\begin{aligned} Var(\bar{X}) &= Var\left(\frac{1}{5} \sum_{i=1}^{5} X_i\right) \\ &= \frac{1}{5^2} Var\left(\sum_{i=1}^{5} X_i\right) \\ &= \frac{1}{5^2} \cdot 5\sigma^2 \\ &= \frac{1}{5}\sigma^2 \end{aligned}$$

3. **Using R as a calculator**

```
library(MASS) ##load the MASS package to your current session
data(Animals)  ##load the Animals dataset into your current session
str(Animals)  ##View the number and types of variables contained in the dataset.
```

```
## 'data.frame':    28 obs. of  2 variables:
##  $ body : num  1.35 465 36.33 27.66 1.04 ...
##  $ brain: num  8.1 423 119.5 115 5.5 ...
```

```
Animals       ##View the dataset in the console window
```

```
##                       body  brain
## Mountain beaver      1.350    8.1
## Cow                465.000  423.0
## Grey wolf           36.330  119.5
## Goat                27.660  115.0
## Guinea pig           1.040    5.5
## Dipliodocus      11700.000   50.0
## Asian elephant    2547.000 4603.0
## Donkey             187.100  419.0
## Horse              521.000  655.0
## Potar monkey        10.000  115.0
## Cat                  3.300   25.6
## Giraffe            529.000  680.0
## Gorilla            207.000  406.0
## Human               62.000 1320.0
## African elephant  6654.000 5712.0
## Triceratops       9400.000   70.0
## Rhesus monkey        6.800  179.0
## Kangaroo            35.000   56.0
## Golden hamster       0.120    1.0
## Mouse                0.023    0.4
## Rabbit               2.500   12.1
## Sheep               55.500  175.0
## Jaguar             100.000  157.0
## Chimpanzee          52.160  440.0
## Rat                  0.280    1.9
## Brachiosaurus    87000.000  154.5
## Mole                 0.122    3.0
## Pig                192.000  180.0
```

Let Y_i be the brain weight of species i. Type the following command to copy the brain weights from the dataset into a vector object y:

```
y<-Animals$brain
```

Using the vector object y, calculate the following:

a) $\sum_{i=1}^{10} Y_i$

```
sum(y[1:10])
```

```
## [1] 6513.1
```

b) $\sum_{i=1}^{10} Y_i^2$

```
sum((y[1:10])^2)
```

```
## [1] 22014450
```

c) $\sum_{i=1}^{28} (Y_i - \bar{Y})^2$

```
sum((y-mean(y))^2)
```

```
## [1] 48114970
```

d) $\sum_{i=1}^{28} Y_i^2 - 28(\bar{Y})^2$

```
sum(y^2)-28*(mean(y))^2
```

```
## [1] 48114970
```

e) The total brain weight for the three dinosaur species: *Dipliodocus*, *Triceratops*, *Brachiosaurus*

```
sum(y[c(6,16,26)])
```

```
## [1] 274.5
```

```
f) The mean (brain weight)/(body weight) ratio over all species.
```

```
mean(Animals$brain/Animals$body)
```

```
## [1] 6.199137
```

4. **Plotting graphs in R**

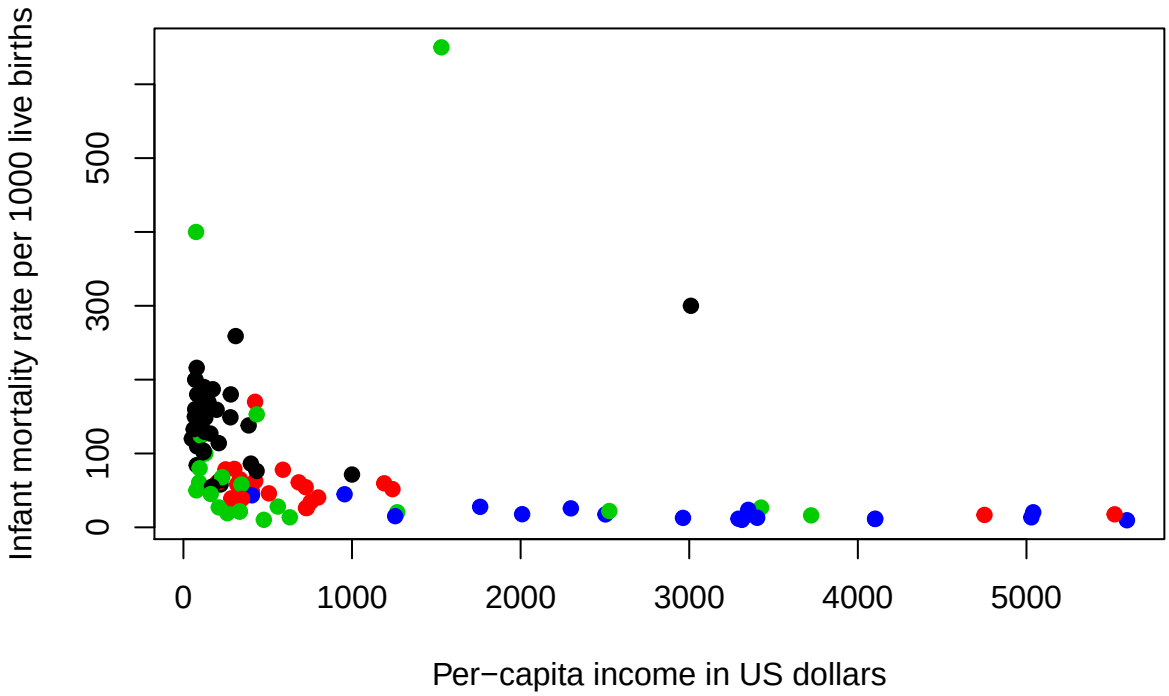
(a) **Graphs using commands from base R installation**

```
world<-read.csv("world1970.csv", header=T) ##read dataset into R environment
str(world) ## get summary of dataset contents
```

```
## 'data.frame':    105 obs. of  5 variables:
##  $ X     : Factor w/ 105 levels "Afganistan","Algeria",..: 4 5 7 15 23 29 30 101 41 43 ...
##  $ income: int  3426 3350 3346 4751 5029 3312 3403 5040 2009 2298 ...
##  $ infant: num  26.7 23.7 17 16.8 13.5 10.1 12.9 20.4 17.8 25.7 ...
##  $ region: Factor w/ 4 levels "Africa","Americas",..: 3 4 4 2 4 4 4 4 4 4 ...
##  $ oil   : Factor w/ 2 levels "no","yes": 1 1 1 1 1 1 1 1 1 1 ...
```

```
(i)
```

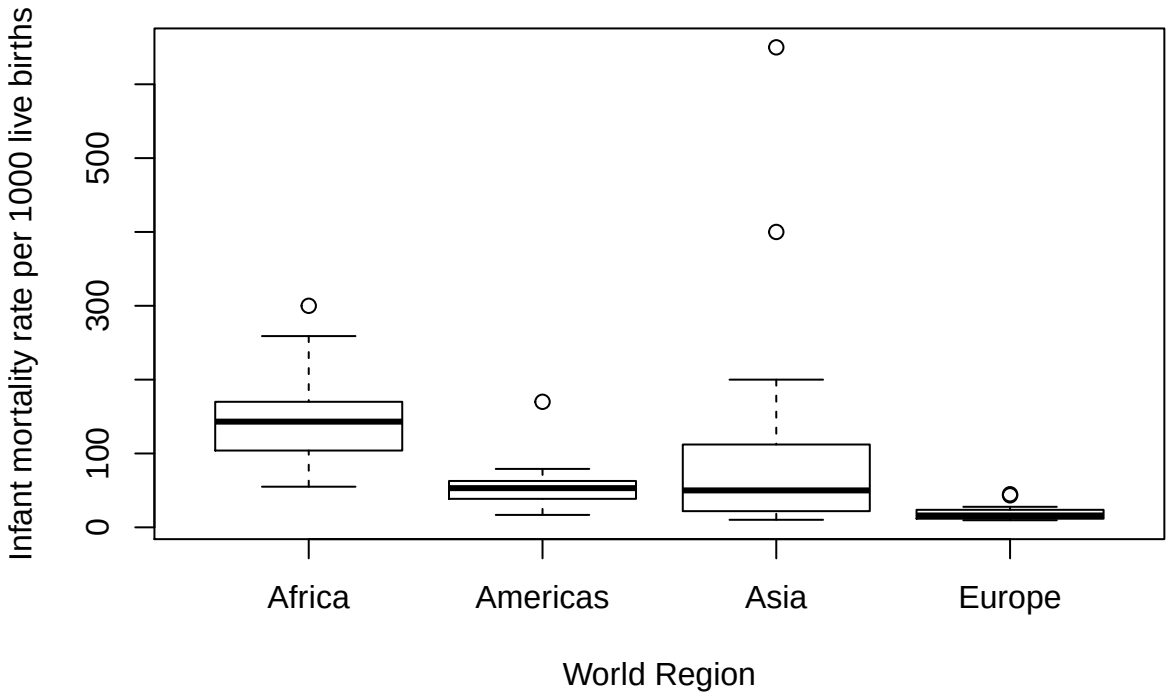
```
    plot(infant ~ income, data=world, xlab="Per-capita income in US dollars",
         ylab="Infant mortality rate per 1000 live births", col=region, pch=19)
```



In general infant mortality decreases as per-capita income increases.

(ii)

```
boxplot(infant ~ region, data=world, xlab="World Region",
        ylab="Infant mortality rate per 1000 live births")
```



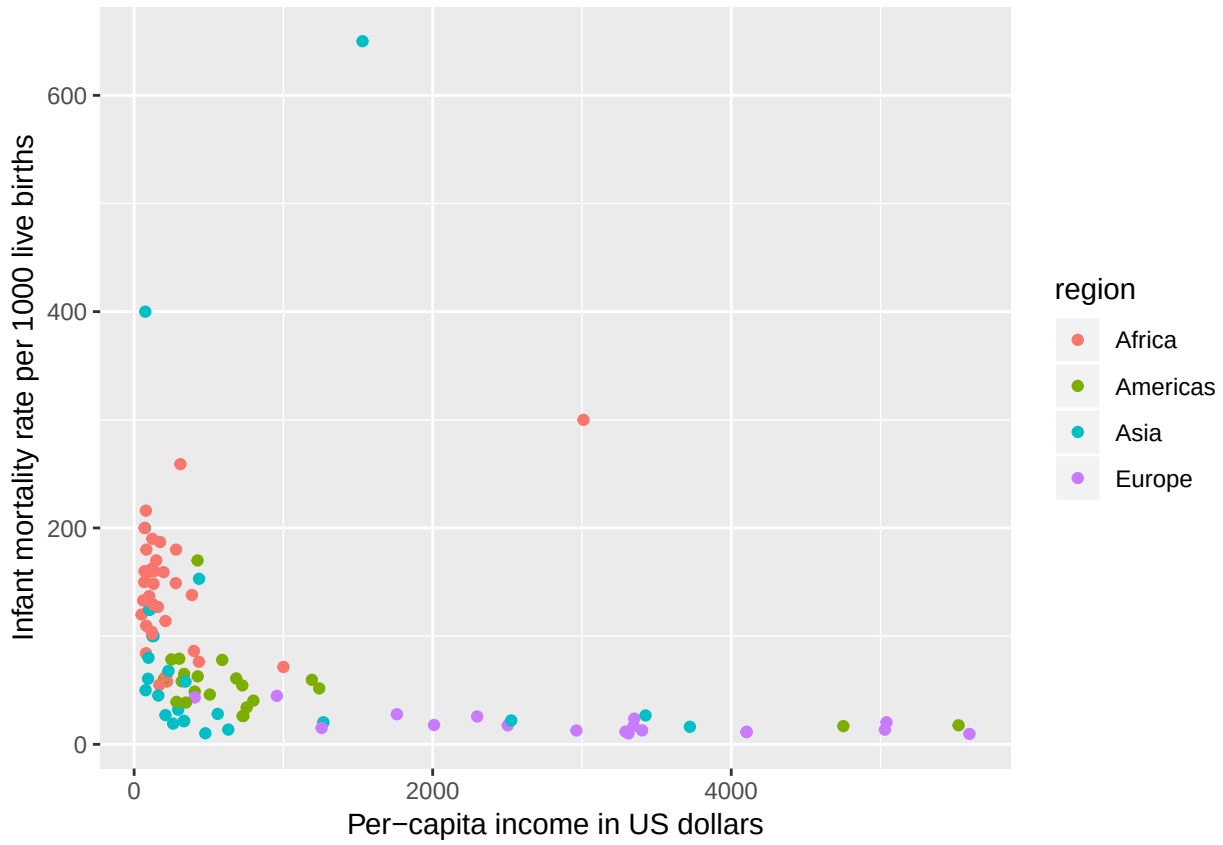
Median infant mortality rate is highest among countries in Africa and lowest in countries in Europe.

(b) **Graphs using commands from `ggplot2` package**

(i)

```
library(ggplot2)
ggplot(world, aes(x=income, y=infant, color=region))+
  geom_point()+
  labs(x="Per-capita income in US dollars", y="Infant mortality rate per 1000 live births")
```

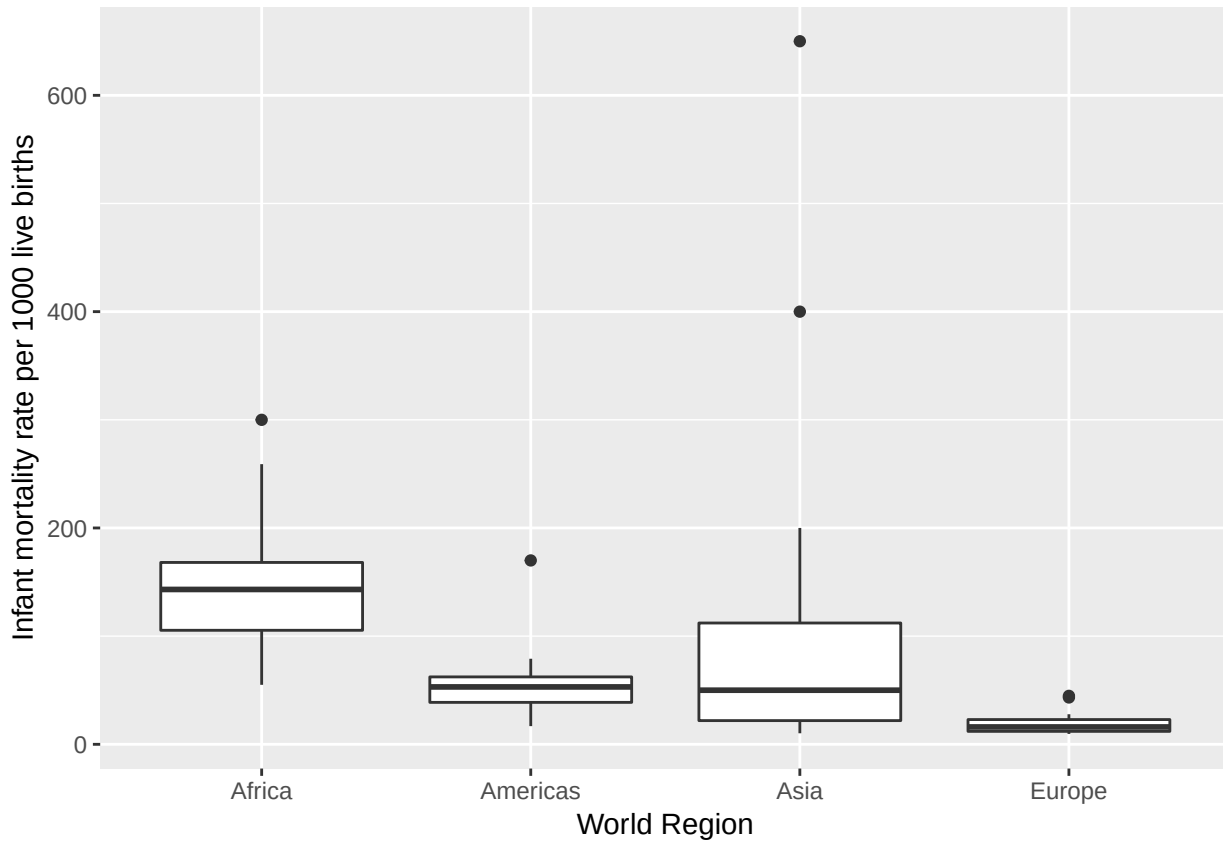
```
## Warning: Removed 4 rows containing missing values (geom_point).
```



(ii)

```
ggplot(world, aes(x=region, y=infant))+
  geom_boxplot()+
  labs(x="World Region", y="Infant mortality rate per 1000 live births")
```

```
## Warning: Removed 4 rows containing non-finite values (stat_boxplot).
```



5. **Obtaining summary statistics in R**:

(a)

```
inc.mean<-mean(world$income)
round(inc.mean,2)
```

```
## [1] 998.07
```

(b)

```
inc.oil<-tapply(world$income, world$oil,mean)
round(inc.oil,2)
```

```
##      no     yes
## 1002.05  955.56
```

(c)

```
regions<-table(world$region) #1-way frequency table
regions #no. of countries in each region
```

```
##
##   Africa Americas     Asia   Europe
##       34       23       30       18
```

```
prop.table(regions) #proportion of countries in each region
```

```
##
##    Africa  Americas      Asia    Europe
## 0.3238095 0.2190476 0.2857143 0.1714286
```

(d)

```r
regions.oil<-table(world$region, world$oil) #2-way frequency table
regions.oil #no. of countries in each region that do or don't export oil
```

```
##
##            no yes
##   Africa   31   3
##   Americas 21   2
##   Asia     26   4
##   Europe   18   0
```

```r
#proportion of countries in each region that do or don't export oil. 1 indicates row proportions
prop.table(regions.oil,1)
```

```
##
##                    no        yes
##   Africa   0.91176471 0.08823529
##   Americas 0.91304348 0.08695652
##   Asia     0.86666667 0.13333333
##   Europe   1.00000000 0.00000000
```